

Software Requirements Specification (SRS)

Project Title: Pakistan Toshakhana — Exploratory Data Analysis (EDA)

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1. Introduction

1.1 Purpose

The purpose of this project is to perform Exploratory Data Analysis (EDA) on the Pakistan Toshakhana dataset. The project will analyze historical records of gifts received by Pakistani public officials and related information available in the dataset. The analysis will identify patterns, trends, missing information, data inconsistencies, and relationships between different attributes.

1.2 Scope

The project includes importing the dataset, understanding its structure, cleaning and preprocessing data, handling missing and duplicate values, identifying inconsistencies, performing statistical analysis, analyzing gift values and relevant attributes, identifying trends and patterns, creating visualizations, and generating insights. The project is limited to exploratory analysis and does not include predictive machine learning.

1.3 Intended Audience

Students, researchers, data analysts, data science learners, and users interested in analyzing publicly available Toshakhana-related records.

2. Overall Description

2.1 Product Perspective

The project is a data analysis system that takes the Toshakhana dataset as input and processes it using Python-based data analysis tools. Workflow: Dataset → Data Cleaning → Data Preprocessing → EDA → Visualization → Insights → Final Report.

2.2 Product Functions

The system will load the dataset, inspect its structure, detect missing and duplicate data, clean inconsistent records, convert data types where required, generate descriptive statistics, analyze variables, identify trends and relationships, create charts, and summarize findings.

2.3 User Characteristics

Users should have basic knowledge of data analysis, Python, Pandas, NumPy, data visualization, and basic statistics.

3. Functional Requirements

FR-01 Dataset Loading

The system shall allow the user to load the Pakistan Toshakhana dataset from a supported data format.

FR-02 Dataset Inspection

The system shall display the number of records, columns, column names, data types, sample records, and basic dataset information.

FR-03 Missing Value Analysis

The system shall identify the number and percentage of missing values in each column.

FR-04 Duplicate Data Detection

The system shall identify duplicate records and determine whether they should be removed.

FR-05 Data Cleaning

The system shall handle missing values, remove unnecessary columns where appropriate, remove duplicates, correct inconsistent formats, and convert data types where required.

FR-06 Statistical Analysis

The system shall generate descriptive statistics including mean, median, minimum, maximum, standard deviation, and quartiles for relevant numerical variables.

FR-07 Categorical Data Analysis

The system shall analyze categorical variables such as recipients, gift categories, and other relevant attributes available in the dataset.

FR-08 Gift Value Analysis

The system shall analyze gift values, including highest and lowest values, average values, distributions, and trends.

FR-09 Time-Based Analysis

Where reliable date/year information is available, the system shall analyze the number of gifts, changes in values, and trends over time.

FR-10 Data Visualization

The system shall generate suitable bar charts, histograms, box plots, line charts, heatmaps, and other appropriate visualizations.

FR-11 Relationship Analysis

The system shall analyze relationships between relevant variables using statistical and visual techniques such as correlation and group comparisons.

FR-12 Outlier Detection

The system shall identify potential outliers using techniques such as box plots and the IQR method.

FR-13 Insight Generation

The system shall summarize major trends, unusual values, frequent categories, significant differences, and data quality issues.

FR-14 Final Report

The system shall produce a final EDA report containing the dataset overview, cleaning process, statistical analysis, visualizations, findings, and conclusions.

4. Non-Functional Requirements

NFR-01 Usability

The analysis should be understandable and visualizations should have appropriate titles, labels, and legends.

NFR-02 Performance

The system should process the dataset within a reasonable time on a standard computer.

NFR-03 Reliability

The analysis should produce consistent results when the same cleaned dataset and procedures are used.

NFR-04 Maintainability

Python code should be organized into logical sections so future analysis can be added easily.

NFR-05 Accuracy

Calculations, statistics, and visualizations should accurately represent the underlying dataset.

NFR-06 Compatibility

The project should be executable in Jupyter Notebook, Google Colab, or VS Code with Python.

NFR-07 Reproducibility

The complete EDA process should be reproducible by executing the notebook or scripts in sequence.

5. System Requirements

5.1 Hardware Requirements

Minimum: Intel Core i3 or equivalent processor, 4 GB RAM, and at least 2 GB available storage.
Recommended: Intel Core i5 or equivalent, 8 GB or more RAM, and at least 5 GB available storage.

5.2 Software Requirements

Windows, Linux, or macOS; Python 3.x; Jupyter Notebook or Google Colab; VS Code (optional). Libraries may include Pandas, NumPy, Matplotlib, Seaborn, and SciPy where required.

6. Data Requirements

6.1 Input Data

The primary input will be the Pakistan Toshakhana dataset obtained from Kaggle.

6.2 Data Preprocessing Requirements

The dataset should be checked for missing values, duplicate records, incorrect data types, inconsistent naming, invalid numerical values, date formatting problems, and unnecessary columns.

7. Exploratory Data Analysis Requirements

7.1 Univariate Analysis

Analysis of individual variables, including numerical distributions, categorical frequencies, minimum and maximum values, and central tendency.

7.2 Bivariate Analysis

Analysis of relationships such as year versus number of gifts, category versus gift value, and recipient versus gift value.

7.3 Multivariate Analysis

Where applicable, multiple variables will be analyzed together to identify more complex relationships and patterns.

7.4 Outlier Analysis

Potential unusual observations will be identified and investigated to determine whether they represent genuine observations, data-entry errors, or exceptional cases.

8. Visualization Requirements

Overview

The project shall use appropriate visualizations to communicate analytical findings.

Analysis	Visualization
Numerical distribution	Histogram
Outliers	Box Plot
Category frequency	Bar Chart
Yearly trends	Line Chart
Correlation	Heatmap
Proportions	Pie Chart

9. Constraints

Constraints

The quality of the analysis depends on the quality of the source dataset. Missing or incomplete records may affect conclusions. Some fields may contain inconsistent information. The project does not verify the authenticity of every individual record and is intended for exploratory analysis rather than legal or official verification.

10. Assumptions

Assumptions

The dataset is accessible and readable; cleaning rules are applied consistently; numerical values are interpreted according to their available definitions; missing information is clearly identified; and EDA findings represent patterns in the dataset rather than causal relationships.

11. Expected Results

Expected Results

The project is expected to provide a cleaned dataset, dataset statistics, missing-value and duplicate analysis, numerical and categorical analysis, outlier analysis, relevant visualizations, trends, relationships, meaningful insights, and a final EDA report.

12. Conclusion

Conclusion

The Pakistan Toshakhana EDA project will provide a structured analysis of the available Toshakhana dataset using data preprocessing, statistical analysis, and visualization techniques. The final outcome will be an organized and reproducible EDA project demonstrating the practical application of Python and data analysis to a real-world dataset.